

THE SHAPE OF A STIMULUS

Trajectory Geometry of Mouse Neuron Population Activity

30563 MATHEMATICAL MODELLING FOR NEUROSCIENCE

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A mouse watches three videos.

**Do its neurons code them as
three distinguishable
population states?**

Why this slice of MICrONS for this research question?



CLIP

Natural movie footage



MONET2

Grey-scale textured noise

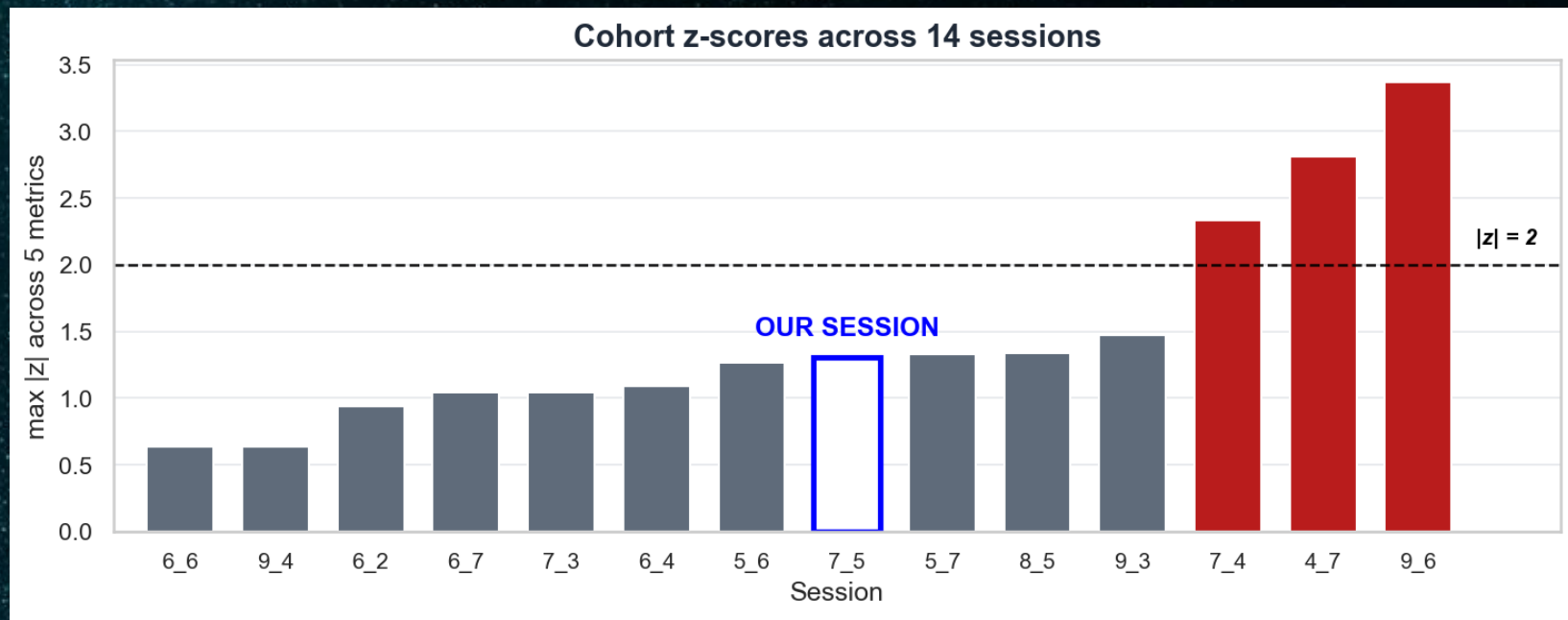


TRIPPY

Synthetic wave patterns

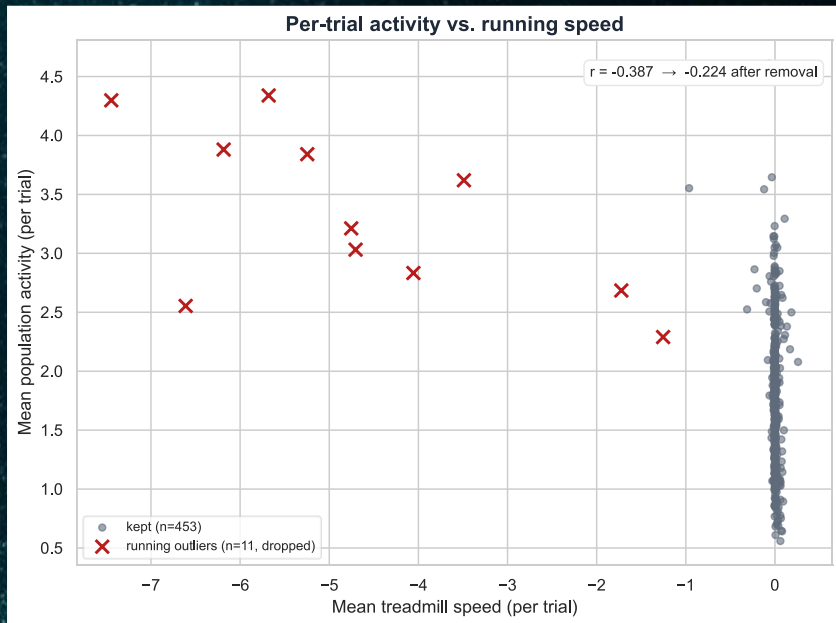
FUNCTIONAL DATA · ~8,000 NEURONS RECORDED SIMULTANEOUSLY
4 CORTICAL AREAS · V1 · LM · RL · AL

Session choice and trial structure

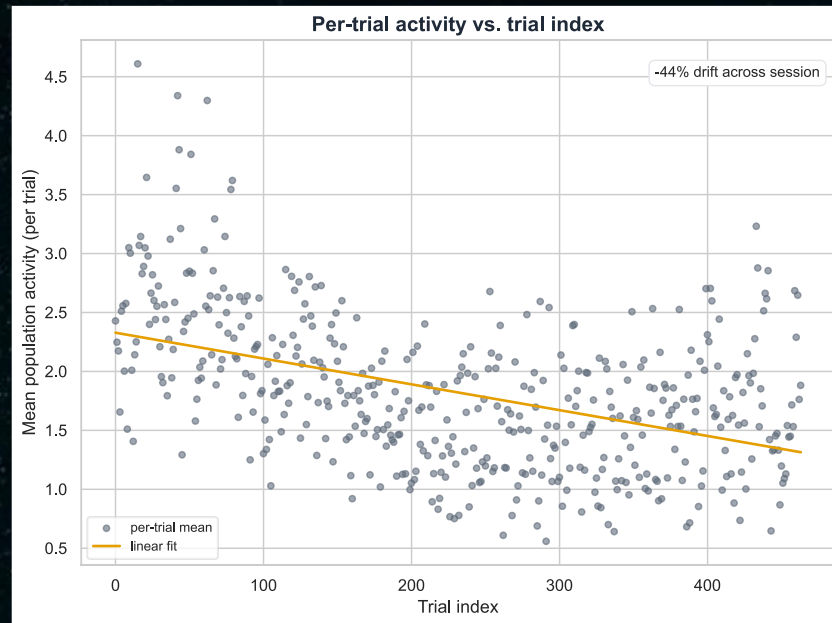


**SESSION 7_5 · $|z| \leq 1.31$ ACROSS ALL METRICS · 8,194 NEURONS (COHORT MEDIAN: 8,176)
453 TRIALS · 377 CLIP · 38 MONET2 · 38 TRIPPY · INTERLEAVED**

Preprocessing techniques



Running influences V1 — 11 outlier trials removed (2.4%)

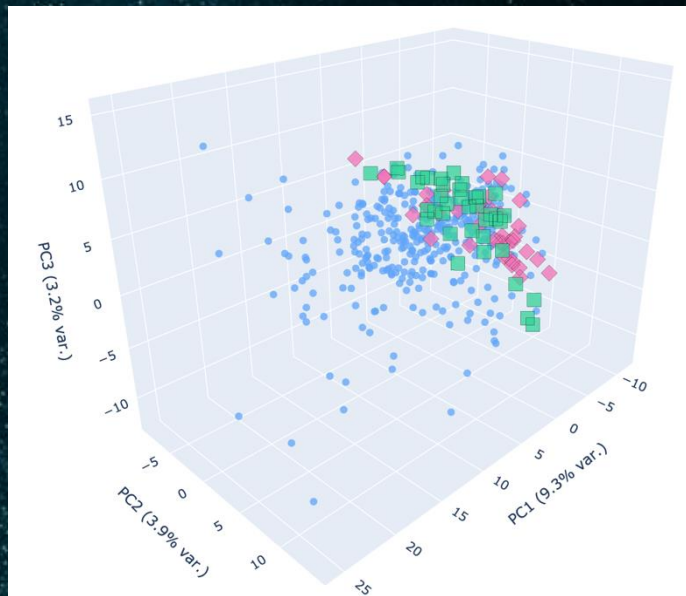


Photobleaching — linear detrend per trial

PIPELINE: OUTLIER FILTER → LINEAR DETREND → PER-NEURON Z-SCORE

Experiment 1: trial-mean PCA

n = 453 trials · 5,485 neurons · top 3 PCs



PCA does not encode any signal

BUT

98.9%

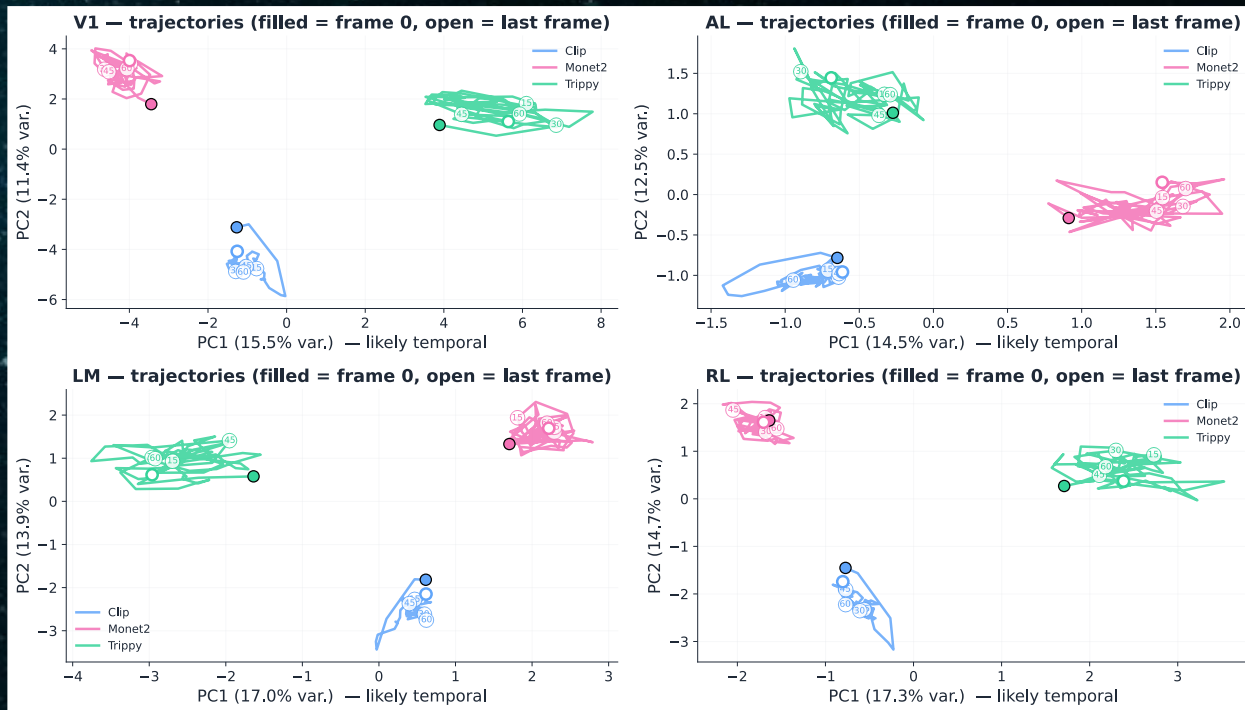
classifier accuracy on the same trial-mean vectors
majority baseline: 83.2%

LM 98.0% · RL 97.8% · AL 94.7%

THE SIGNAL IS THERE. PCA ISN'T ALIGNED WITH IT.

Experiment 2:

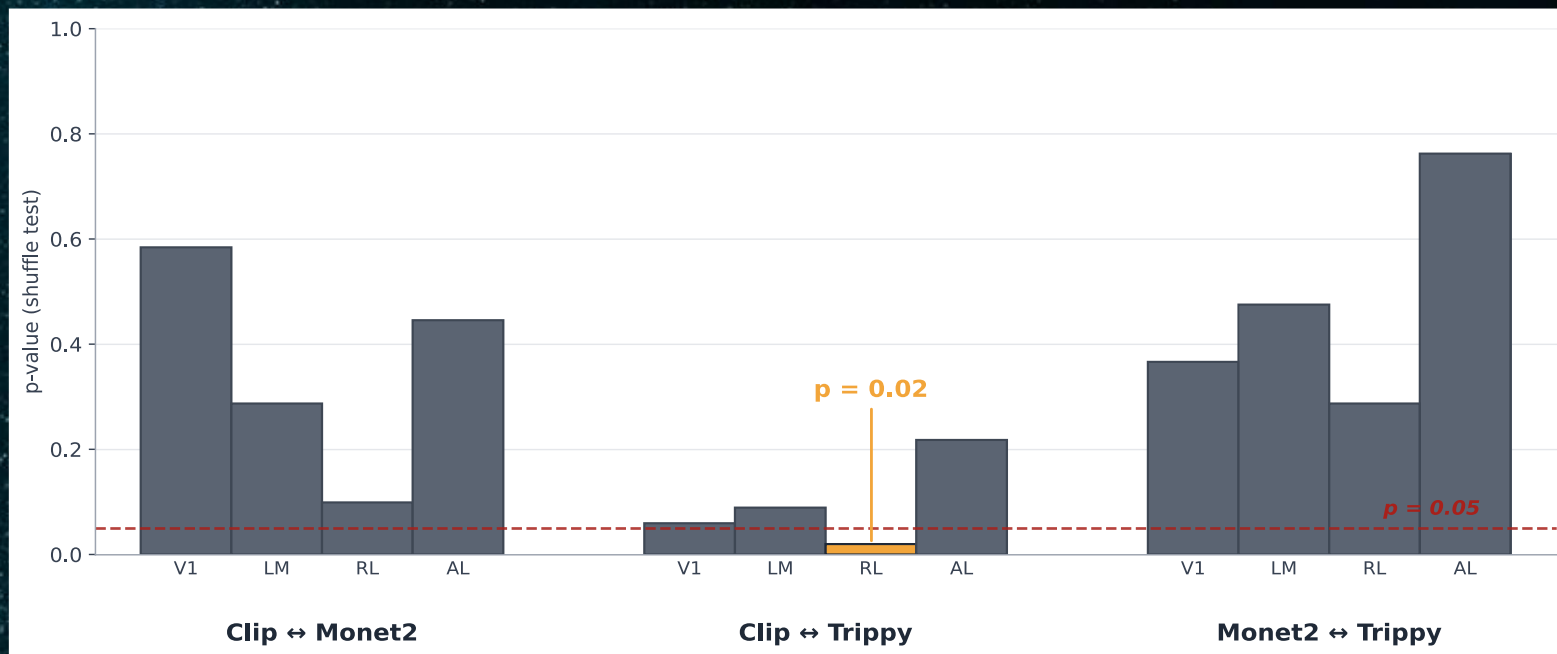
Trial-averaged trajectories - projected into PC space



Three separated clusters in every area
BUT IS THE SEPARATION REAL?

Biased Test: PCA fit on real-label trial averages, then shuffled data projected through the same PCs

Honest Test: full feature space with no projection on all neurons



Monet2 ↔ Trippy never reliably separates in any area

THE COLLAPSE PERSISTS

Two controls - both pass

TRIAL-COUNT CONTROL

THE ISSUE

Clip has 377 trials —
Monet2 and Trippy only 38

WHAT WE DID

Subsampled Clip down to 38 trials.

RESULTS

Distances unchanged

Trial-count is not the explanation.

NEURON-COUNT CONTROL

THE ISSUE

V1 has 5,485 neurons —
AL only 414.

WHAT WE DID

Matched all areas to 414 neurons.

RESULTS

V1's lead vanishes.

All areas sit at the same level.

LINEAR METHODS HIT A CEILING

Experiment 3: same method—zoomed inside Clip



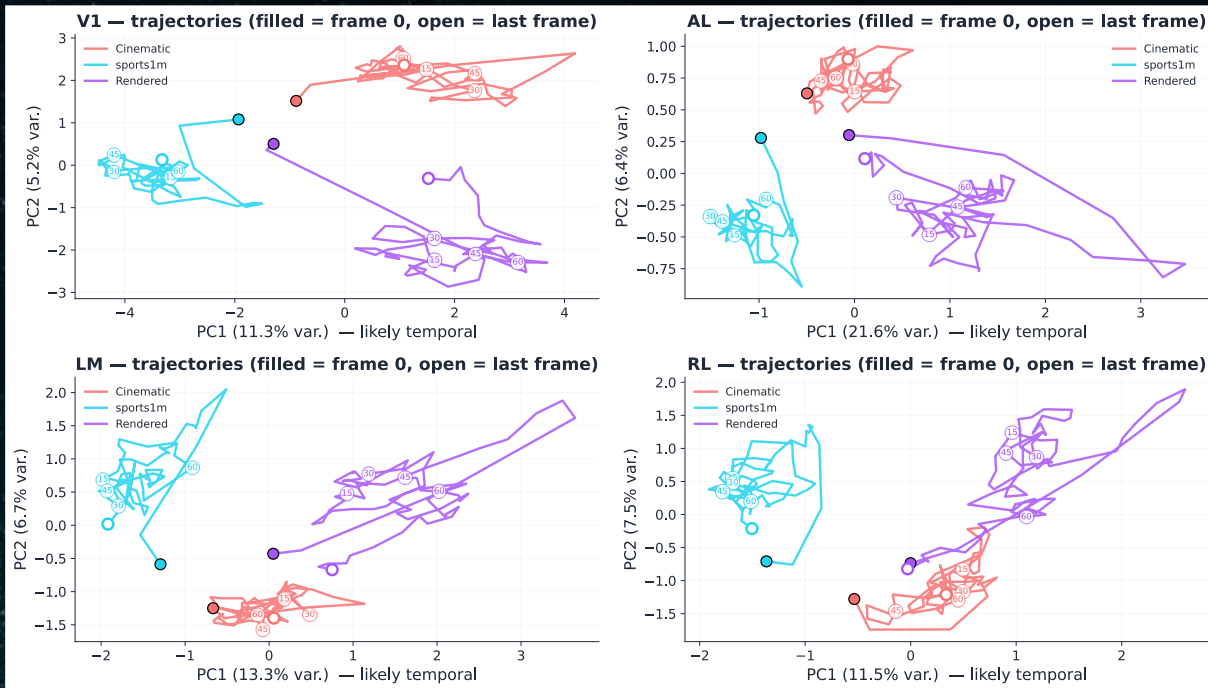
Cinematic



sport1m



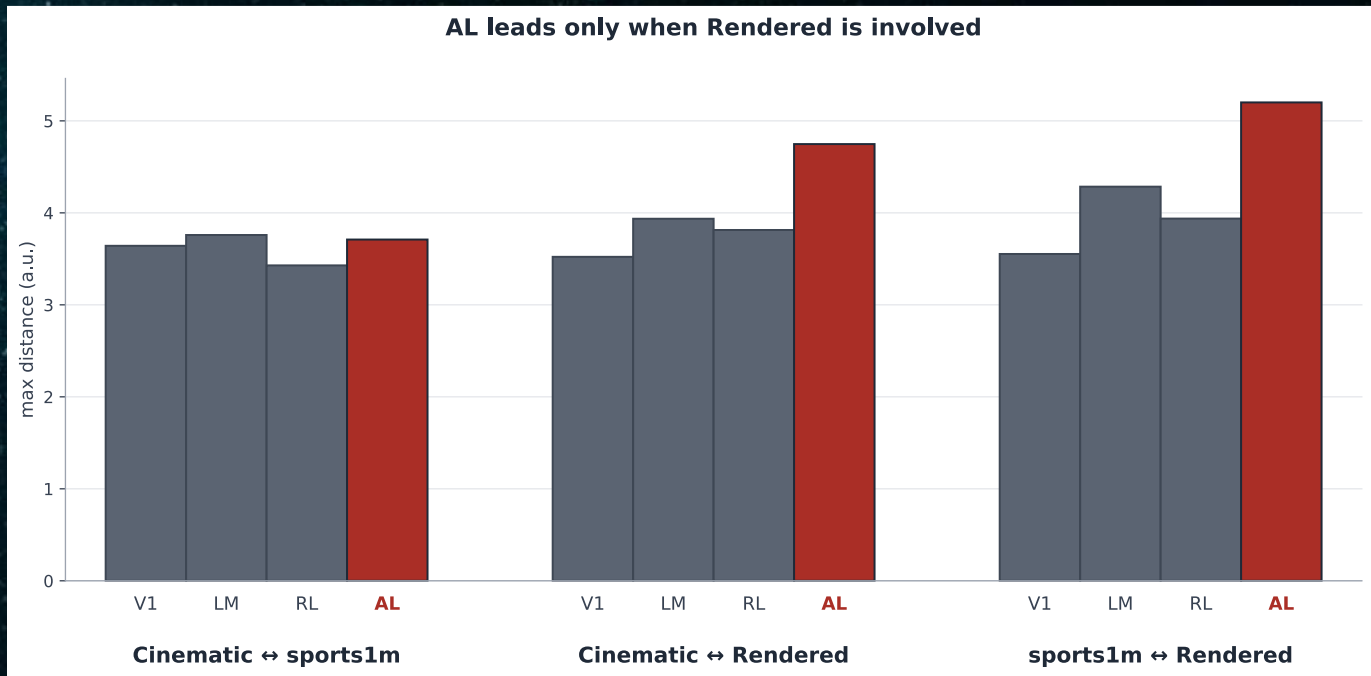
Rendered



Three sub-categories separate in every area with $p \leq 0.02$

BUT IS ONE AREA SPECIAL?

After population matching – per-neuron signal



AL is a naturalness detector, the same axis Experiment 2 found.

AL DETECTS ARTIFICIALITY, NOT THE CATEGORY.

The same boundary, again.

LIMITATIONS

Linear methods may miss
non-linear structure

Single session (7_5) —
cohort generalization untested

Connectomics unused —
structural link untested

WHAT'S NEXT

CEBRA or manifold embeddings
designed for exactly this regime

Cross-session replication

Functional $\Leftrightarrow$ **structural** link
*use the connectome to test whether functional
clusters reflect wiring*

**Linear methods sense the naturalness boundary
but hit a ceiling for synthetic vs. synthetic**

Any questions?

THANK YOU!